

Medicine

Departmental Objectives

At the end of the course, the students should be able to:

- Mention the various manifestations of common diseases.
- Mention the basic principle of history taking and clinical examinations.
- Elicit required history; perform a physical examination including examination of an unconscious patient and patient of syncope.
- Correlate the clinical symptoms and physical signs to make a provisional diagnosis of common diseases and suggest relevant investigation & Interpret reasonably.
- Initiate initial management of commonly encountered health problem.
- Diagnose and refer acute medical emergencies like acute myocardial infarction, acute pulmonary oedema acute anaphylactic and hypovolumic shock, status asthmaticus, tension pneumothorax, hyperpyrexia, haemoptysis, gastro-intestinal bleeding, diabetic coma, common poisoning etc.
- Acquire the skills to perform minor procedure like – IV cannulation, insertion of nasogastric tube, urinary bladder catheterization etc.
- Demonstrate communication skills to establish effective communications with patients and with scientific community.
- Be familiar with various national health programs.
- Practice medical ethics in service delivery.
- Mention the prophylaxis of common infectious diseases.
- Interpret common haematological report, x-ray chest, ECG, ultra sonogram and CT scan of brain.

List of Competencies to acquire:

- History taking, communication skill, clinical examination.
- Diagnosis of common clinical problems.
- Able to record the arterial pulse, blood pressure, temperature and respiration.
- Insertion of a cannula
- Insertion of Ryle's tube
- Catheterization
- Drawing of blood
- Writing a BT order and transfusion note.
- Identification of instruments used in medicine
- Able to interpret common X rays, ECG, ultra sonogram and CT scan of brain.
- Perform CPR.

Distribution of Teaching - Learning Hours

Lecture	Tutorial	Practical	Clinical Teaching	Total Teaching hours	Integrated teaching (Common)	Formative Exam		Summative exam	
						Preparatory leave	Exam time	Preparatory leave	Exam time
90 hrs	—	10hrs	144 hrs	244 hrs	10 hrs	10 days	20 days	10 days	25 days

Teaching - Learning Methods, Teaching Aids and Evaluation

Teaching Methods			Teaching aids	In course evaluation
Large group	Small group	Self learning		
Lecture	- Bed side teaching (Problem based learning) - OPD- teaching - Demonstration in wards	- Assignment - Self study	- Laptop - Computer - Multimedia - White board & Marker - Video - X-ray plate - View Box - ECG - Haematological report - CT scan and USG report	- Item Examination - Card final - Term Examination - Term final (written, oral, OSPE, clinical)

Related Equipments/Instrument:

- Foley's catheter
- Plain rubber catheter
- I/V cannula
- Bone marrow aspiration needle
- Lumber puncture needle
- ESR tube
- Ryle's tube
- Tongue depressor.
- Syringe
- I/V fluid ,infusion & transfusion set.
- ORS

Professional Examination:

Marks distribution of Assessment : total marks = 300

- Written = 100 (SAQ=70 + MCQ=20+ 10 formative assessment)
- SOE = 100 (two boards 4 examiners, 50 marks in each board)
- Clinical + practical = 100 (long case-40, short case-15x2= 30, OSPE-30)

Learning Objectives and Course Contents in Medicine

Lecture – 90 hrs

Learning Objectives	Contents	Teaching Hours
Students should be able to – ● Mention the principles of medical ethics ● Describe the clinical features common genetic diseases ● Mention the investigations required for common genetic diseases ● Counsel about common genetic disorder, e.g. Hemophilia, thalassemia.	General ● Principles of medical ethics ● Clinical genetics - common types, investigation ● Prevention of genetic diseases and genetic counseling.	L - 2
Students should be able to – ● Assess nutritional status properly ● Evaluate nutritional problem from the history. ● Assess nutritional ailments, by physical examination. ● Interpret & correlate physical findings with investigation. ● Assess BMI	Nutritional and metabolic disorders ● Nutritional assessment & needs ● Protein energy malnutrition ● Obesity ● Vitamin and mineral deficiency & excess	L - 3
Students should be able to – ● Diagnose water and electrolyte imbalance ● Diagnose acid based disorder. ● Interpret investigation report e.g. Serum electrolyte, ABG ● Manage emergencies	Water, electrolyte and acid-base imbalance ● Acid-base disorders ● Fluid and electrolyte disturbances	L - 2

Learning Objectives	Contents	Teaching Hours
Students should be able to – ● Diagnose critically ill patient ● Provide initial emergency management and proper referral.	Critical care Medicine ● Major manifestations of critical illness ● Circulatory failure: shock ● Respiratory failure ● Renal failure ● Coma ● Sepsis ● Disseminated intravascular coagulation ● General principles of critical care management	L -3
Students should be able to – ● Classify psychiatric illness	Medical Psychiatry ● Classification of psychiatric disorder and evaluation. ● Symptoms analysis	L -2
Students should be able to – ● Diagnose common poisoning ● Manage emergency with proper referral	Poisonings ● General approach to the poisoned patient ● Poisoning by specific pharmaceutical agent, chemicals and pesticides.	L - 3
Students should be able to – ● Describe environmental occupational hazard. ● Manage emergencies	Specific environmental and occupational hazard ● Heatstroke and hypothermia	L -1

Learning Objectives	Contents	Teaching Hours
Students should be able to – ● Diagnose infectious disease from history and physical findings. ● Interpret and co-relate physical findings with investigation, e.g. hematology, immunological investigation, x-ray, stool and urine examination and sputum examination. ● Manage common infectious disease. ● Provide emergency management and proper referral.	Infection ● Prevention of infectious disease. ● Hospital acquired infections ● Fever of unknown origin ● Typhoid and paratyphoid fevers ● Leprosy ● Acute gastroenteritis & food poisoning ● Common exanthemata ○ Measles ○ Mumps ○ Rubella ○ Varicella ● Common viral respiratory infections, influenza. ● Dengue fever ● Sexually transmitted disease.	L -10
Students should be able to – ● Evaluate cardiovascular disease from history ● Perform proper examination of cardiovascular system ● Assess cardiovascular ailments by physical examination ● Interpret and co-relate physical findings with investigational report e.g. X-ray chest, hematology and ECG. ● Provide initial emergency management and proper referral.	Cardiovascular system ● Major manifestations of cardiovascular disease ● Hypertension ● Heart failure ● Ischaemic heart disease, cardiac arrest. ● Disorders of heart rate, rhythm and conduction. ● Infective endocarditis. ● Rheumatic fever ,valvular heart disease. ● Congenital heart disease.	L -10

Learning Objectives	Contents	Teaching Hours
● Evaluate respiratory problem from the history. ● Perform examination of respiratory system. ● Assess respiratory ailments, by physical examination ● Interpret investigation reports e.g. X-ray chest, sputum examination, MT test and hematology. ● Provide emergency management and proper referral.	Respiratory system ● Major manifestations of lung disease. ● Pneumonia. ● Bronchial asthma ● Chronic obstructive pulmonary disease ● Pulmonary tuberculosis ● Pneumothorax ● Bronchogenic carcinoma	L -8
● Evaluate renal problem from the history. ● Perform examination of renal system. ● Assess renal ailments, by physical examination. ● Interpret and co-relate physical findings with investigation findings e.g. Urine examination and hematology. ● Provide emergency management and proper referral.	Kidney and genitourinary system ● Major manifestations of renal and urinary tract disease ● Acute and chronic renal failure ● Infections of the kidney and urinary tract	L -5
● Evaluate G.I.T. problem from the history ● Perform examination of G.I. system ● Assess G.I.T. ailments, by physical examination ● Interpret & Co-relate physical findings with investigation, e.g. Hematology, ultrasonography, stool examination, ascitic fluid study. ● Provide emergency management and proper referral.	Gastrointestinal system ● Major manifestations of gastrointestinal disease ● Diseases of the stomach and duodenum - gastritis, peptic ulcer disease ,non ulcer dyspepsia. ● Intestinal tuberculosis ● Malabsorption syndrome ● G.I. bleeding. ● G.I. malignancy.	L - 5

Learning Objectives	Contents	Teaching Hours
● Evaluate liver disease from history ● Perform examination of hepatobiliary system ● Diagnose common diseases of hepatobiliary system ● interpret common investigations findings ● Provide initial management of hepatobiliary emergencies with referral	Endocrinology and Metabolism ● Diabetes mellitus ● Hyperthyroidism ● Hypothyroidism ● Cushing syndrome. ● Adrenal insufficiency ● Acromegaly	L -6
● Evaluate endocrine and metabolic problem from the history. ● Assess endocrine ailments, by physical examination. ● Diagnose common endocrine disease ● Interpret endocrine investigation findings , e.g. biochemistry & hormone analysis. ● Provide initial management of endocrine emergencies with referral	Endocrinology and Metabolism ● Diabetes mellitus ● Hyperthyroidism ● Hypothyroidism ● Cushing syndrome. ● Adrenal insufficiency ● Acromegaly	L -7
● Evaluate Haemopoetic problem from the history. ● Assess Haemopoetic ailments, by physical examination ● Interpret and co-relate physical findings with disease ● Interpret common hematological investigation findings, e.g. CBC, PBF, bone marrow study	Hematological disorders ● Major manifestations of hematological diseases ● Anaemia ● Thrombocytopenia ,ITP. ● Pancytopenia ● Haematological malignancies ● Lymphoma ● Bleeding & coagulation disorder.	L -7

Learning Objectives	Contents	Teaching Hours
● Suspect immune deficiency ● Identify hyper sensitivity reaction ● Provide emergency management of anaphylactic shock with proper referral	Disorders of the immune system ● Introduction to the immune system and autoimmunity ● Hyper sensitivity reaction , anaphylaxis ● HIV, AIDS and related disorders	L - 3
● Diagnose common skin diseases. ● Provide initial management	Skin diseases ● Major manifestations of skin disease ● Urticaria ● Scabies ● Fungal infections ● Eczema	L -2
● Evaluate neurological problem from the history. ● Perform examination of nervous system with interpretation of common physical findings. ● Diagnose common neurological disease. ● Interpret common investigations findings , e.g. CSF study, CT scan ● Provide initial management of neurological emergency. ● Refer patient to the proper place.	Neurological diseases ● Headache and facial pain ● Disease of cranial nerves ● Cerebrovascular disease ● Meningitis, Encephalitis. ● Seizure disorder. ● Diseases of spinal cord- cervical spondilitis. ● Peripheral neuropathy,GBS.	L -8
● Evaluate connective tissue disorder from the history. ● Diagnose common connective tissue disease. ● Interpret common investigations findings.	Connective tissue and joints ● Rheumatoid arthritis ● Ankylosing spondyliti.	L -3

- **Clinical teaching**
 - **Inpatient department -110 hrs**
 - **Outpatient department -24 hrs**
- **Practical**
 - **10 hrs**

Inpatient Department

Learning Objectives	Contents	Teaching Hours
History taking procedure	● Chief complain and elaboration of symptoms ● History of present and past illness ● Family history ● Personal history ● Drug history, etc	10
Perform general physical examination	● Anemia, jaundice, cyanosis, clubbing, koilonychias etc ● Blood pressure, pulse, JVP, temperature etc.	6
Analyze symptoms of GI system	● Symptoms of GI system- abdominal pain, vomiting, diarrhea, hematemesis, malena, dyspepsia etc	10
Perform examination of GI system	● Inspection, palpation, percussion and auscultation of abdomen	8
Analyze symptoms of respiratory system	● Cough, SOB, chest pain, haemoptysis, sputum, fever etc	6
Perform examination of respiratory system	● Inspection, palpation, percussion and auscultation of chest	12
Analyze symptoms of cardiovascular system	● Angina, SOB, palpitation, oedema etc	8
Perform examination of cardiovascular system	● Inspection, palpation and auscultation of precordium	10
Analyze symptoms of nervous system	● Headache, unconsciousness, cranial nerve palsy, convulsion, paralysis etc	8
Perform examination of nervous system	● Cranial nerve, motor examination, sensory, cerebellar examination	10
Analyze symptoms of renal system	● Anuria, oliguria, polyuria etc	6
Perform examination of renal system	● Kidney palpation	4
Analyze symptoms of musculoskeletal system	● Arthritis, arthralgia, muscle weakness	6
Perform examination of musculoskeletal system	● Examination of joints	6

Outpatient department ----- Teaching time—24 hrs.

- Diagnose common disease
- Practice different physical examinations
- Practice prescription writing
- Practice patient counseling

Practical

Learning Objectives	Contents	Teaching Hours
Identify common instruments, X- ray, ECG, CT scan,USG report etc and their use	● Bone marrow needle ● LP needle ● Ryle's tube ● Folly's catheter ● I/V fluid ● ESR tube ● Syringe ● X- ray chest, abdomen ● ECG.USG abdomen. ● Tongue depressor.	10